

On LLR Calculations for Soft-decision Decoding in Next-generation IM-DD Systems with Laser RIN

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Abstract We propose a low-complexity LLR approximation for SD decoding in IM-DD systems with relative intensity noise. Our approximation results in no BER performance penalty for a concatenated KP4+Hamming FEC scheme, and outperforms mismatched AWGN-based LLRs. ©2026 The Author(s)

Introduction

The rapid growth of artificial intelligence applications is driving demand for higher speeds in short-reach data center intraconnects (DCI) [1]. Low hardware costs and low power consumption are essential, and thus, DCI optical transceivers employ intensity-modulation (IM) and direct-detection (DD) with 4-ary pulse amplitude modulation (PAM-4) [2]. Scaling the data rates to 400 Gb/s/lane and beyond faces significant challenges due to the bandwidth constraints of electro-optical components and noise impairments [3, 4]. One important noise impairment in such systems is relative intensity noise (RIN) arising from the laser source.

To achieve the scaling to 400 Gb/s per lane and beyond, several technologies have been studied, such as advanced digital signal processing (DSP) techniques [5] or forward error correction (FEC) codes with low code rates [6]. To improve the coding gain with respect to KP4, several proposals have been made. For example, a concatenated FEC scheme based on the standard KP4 code [7] as the outer code with hard-decision (HD) decoding, and a Hamming code as the inner code with *soft-decision* (SD) decoding was proposed for the 200 Gb/s/lane baseline in [8].

SD decoding typically uses log-likelihood ratios (LLR). Calculating LLRs requires knowledge of the channel, which is not always straightforward to obtain accurately [9]. Channels with signal-dependent noise, such as RIN, induce different noise statistics per symbol. LLR calculations for PAM systems with signal-dependent noise have been studied in works such as [10], [11] (using

machine learning), and [12] (using adaptive calculation algorithms). For the additive white Gaussian noise (AWGN) channel, LLRs are well approximated by piecewise linear functions of the received values. However, for channels with signal-dependent noise such as RIN, this is no longer the case. Both LLR expressions and low-complexity approximations for PAM-4 in high-speed DCI IM-DD systems with RIN have yet to be presented.

In this paper, we study the LLR calculations of an IM-DD system limited by RIN. First, we show that the LLRs for this channel follow a quadratic polynomial relationship as a function of the received values. We also show that using mismatched AWGN-based LLRs that neglect signal-dependent noise results in a significant penalty in terms of post-FEC bit error rate (BER). Furthermore, we propose a low-complexity piecewise linear approximation of the LLRs and show that there is virtually no post-FEC BER performance loss when using our proposed approximation as the soft information for the SD decoder.

System Model

We consider the concatenated FEC scheme depicted in Fig. 1. KP4-coded bits are encoded with a (128, 120) extended Hamming code, and then modulated into PAM-4 symbols $X \in \mathcal{X} = \{\pm\Delta, \pm3\Delta\}$ using Gray labeling, where Δ is a scaling factor. The symbols X are the input to the standard IM-DD system shown in Fig. 1. The transmitter (TX) DSP applies pulse shaping and nonlinear pre-distortion to compensate for the modulator transfer function. The DAC generates the analog waveform which is then conditioned by the driver

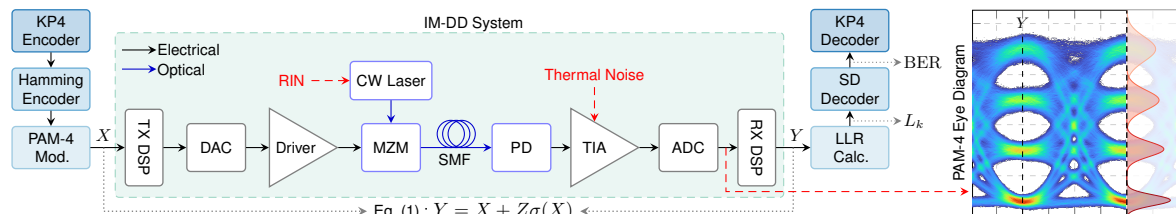


Fig. 1: Considered system: concatenated KP4+Hamming FEC scheme with an IM-DD channel with signal-dependent noise. The eye diagram shows a PAM-4 signal after the ADC, where each PAM level has a different noise distribution due to RIN as shown by the red Gaussians. The LLRs L_k are calculated with either (4), (5), or (6) and (7), and then fed to the SD decoder.

for IM. A Mach-Zehnder modulator (MZM) is used to modulate the intensity of an O-band (1310 nm) continuous-wave (CW) laser with RIN.

The optical signal is transmitted through a single-mode fiber (SMF). Given the short fiber length of < 500 m, both chromatic dispersion and attenuation are negligible. At the receiver (RX), a photodiode (PD) generates current proportional to the incident optical light. The signal bias from IM is removed and a transimpedance amplifier (TIA) converts the current to voltage. During this process, thermal noise is generated.

The analog waveform is sampled by the ADC to be further processed by the receiver DSP. The RX DSP applies matched filtering, downsampling, and equalization to remove intersymbol interference (ISI). For this work, we consider any remaining ISI after equalization to be negligible. Fig. 1 shows an example of an eye diagram for the system under consideration, where the presence of RIN is clearly visible. RIN generates a noise distribution proportional to the square of the symbol's optical amplitude, as depicted by the red Gaussians at the maximum eye opening.

We consider the same channel model as in [13, 14], where the channel output Y is modeled as

$$Y = X + Z\sigma(X), \quad (1)$$

where Z is a zero-mean, and unit-variance Gaussian random variable to model the noise. The total noise variance is modeled with the function $\sigma(X)$, defined as

$$\sigma^2(X) \triangleq p_0 + p_1(X + \beta)^2, \quad (2)$$

where p_0 and p_1 are coefficients that depend on thermal noise and RIN noise statistics, respectively [15, Eq. (10)], [16, Eq. (8)]. Furthermore, the IM bias $\beta = 3\Delta \frac{\text{ER}+1}{\text{ER}-1} \geq |\min\{\mathcal{X}\}|$ models the extinction ratio ER of the system [17, Sec. III].

For the channel model in (1), the conditional probability density function (PDF) of the output is

$$f_{Y|X}(y|x) = \frac{1}{\sqrt{2\pi\sigma^2(x)}} \exp\left(-\frac{(y-x)^2}{2\sigma^2(x)}\right), \quad (3)$$

where $\sigma^2(x)$ is the noise variance from (2) conditioned on each symbol x . The PDF (3) is needed by the LLR calculation block (see Fig. 1). The resulting LLRs are then fed to the SD decoder to decode the Hamming code. In this work, we present results up to the output of the SD decoder. We refer to these BER values as post-FEC BER.

Log-likelihood Ratio Calculations

For the channel output Y with PDF (3), the LLR for the bit position $k = 1, 2$ is calculated as

$$L_k = \log \frac{\sum_{x \in \mathcal{X}_k^1} \frac{1}{\sigma(x)} \exp\left(-\frac{(y-x)^2}{2\sigma^2(x)}\right)}{\sum_{x \in \mathcal{X}_k^0} \frac{1}{\sigma(x)} \exp\left(-\frac{(y-x)^2}{2\sigma^2(x)}\right)}, \quad (4)$$

Tab. 1: Simulation parameters and values for 400 Gb/s

Parameter	Value	[Unit]
Symbol rate	226	[GBd]
Optical modulation amplitude	0	[dBm]
Extinction ratio	4.5	[dB]
Link losses	3.2	[dB]
TIA input-referred noise	22	[pA/ $\sqrt{\text{Hz}}$]
Relative intensity noise	-143	[dB/Hz]
Noise bandwidth	135.5	[GHz]
Constant $\sigma^2(X) : p_0$	$1.219 \cdot 10^{-10}$	[V ²]
Constant $\sigma^2(X) : p_1$	$1.358 \cdot 10^{-3}$	[-]

where $\mathcal{X}_k^b \subset \mathcal{X}$ is the subset of symbols where the bit in position k has a value of $b \in \{0, 1\}$. From (4), it can be seen that L_k depends on $\sigma(x)$.

A receiver that is unaware of RIN would calculate an average variance over all transmitted and received symbols. Let this variance be denoted by $\bar{\sigma}^2$. The LLRs in (4) then become

$$L_k^{\text{awgn}} = \log \frac{\sum_{x \in \mathcal{X}_k^1} \exp\left(-\frac{(y-x)^2}{2\bar{\sigma}^2}\right)}{\sum_{x \in \mathcal{X}_k^0} \exp\left(-\frac{(y-x)^2}{2\bar{\sigma}^2}\right)}. \quad (5)$$

This coincides with the LLR calculation of a linear AWGN channel, hence we denote (5) as L_k^{awgn} .

Figure 2(a) shows LLR calculations for the system in Fig. 1 with the parameters in Tab. 1, using (4) and (5). The results in this figure show that the exact LLRs L_k behave close to a piecewise quadratic function due to the argument $(y-x)^2/2\sigma^2(x)$ in the exponential of (3). On the other hand, the AWGN LLRs L_k^{awgn} behave close to a symmetric piecewise linear function of the output y . Furthermore, it is clear that there are noticeable differences between L_k and L_k^{awgn} , which is expected to lead to a performance penalty in post-FEC BER if the mismatched LLRs L_k^{awgn} are used instead of L_k [9, Sec. V].

We next propose a piecewise linear approximation of (4) that is simpler to compute and, as shown in the next section, does not incur a performance penalty. LLRs with magnitude close to zero represent the least reliable bits which are critical for SD decoding. We therefore propose an approximation around $L_k = 0$, i.e., the so-called zero-crossing (ZC) approximation introduced in [18, Sec. III-C]. To derive expressions and compute the ZC points of L_k and its slope at these points, we use the max-log approximation [19, Eqs. (4)-(6)] to simplify the summations inside the logarithm of (4) to single exponential terms that are easier to handle analytically. For ease of notation, we now define $\sigma_1 \triangleq \sigma(-3\Delta)$, $\sigma_2 \triangleq \sigma(-\Delta)$, $\sigma_3 \triangleq \sigma(\Delta)$, and $\sigma_4 \triangleq \sigma(3\Delta)$, with OMA = $3\Delta - (-3\Delta) = 6\Delta$ being the optical modulation amplitude (OMA).

From Fig. 2(a), we see that L_1 has only one ZC point which is given by $-\Delta(\sigma_3 - \sigma_2)/(\sigma_3 + \sigma_2)$.

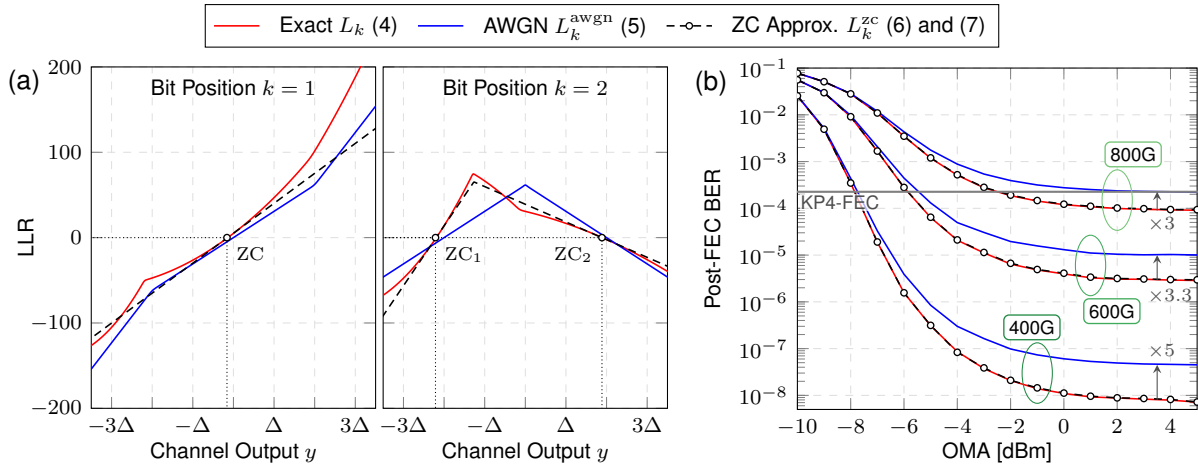


Fig. 2: Comparison of the (a) LLR calculations L_k , L_k^{awgn} , and L_k^{zc} , and (b) the resulting post-FEC BER performance.

The slope at this point is $2\Delta/(\sigma_2\sigma_3)$, and therefore the ZC approximation of L_1^{zc} is

$$L_1^{\text{zc}} = \frac{2\Delta}{\sigma_2\sigma_3} \left(y + \Delta \frac{\sigma_3 - \sigma_2}{\sigma_3 + \sigma_2} \right). \quad (6)$$

Next, we see from Fig. 2(a) that L_2 has two ZC points. The first is at $-\Delta(3\sigma_2 + \sigma_1)/(\sigma_2 + \sigma_1)$ with slope $2\Delta/(\sigma_1\sigma_2)$, while the second is at $\Delta(\sigma_4 + 3\sigma_3)/(\sigma_4 + \sigma_3)$ with slope $-2\Delta/(\sigma_3\sigma_4)$. Since the slopes have opposite sign, the ZC approximation of L_2^{zc} is

$$L_2^{\text{zc}} = \min \left\{ \frac{2\Delta}{\sigma_1\sigma_2} \left(y + \Delta \frac{3\sigma_2 + \sigma_1}{\sigma_2 + \sigma_1} \right), \right. \\ \left. - \frac{2\Delta}{\sigma_3\sigma_4} \left(y - \Delta \frac{\sigma_4 + 3\sigma_3}{\sigma_4 + \sigma_3} \right) \right\}. \quad (7)$$

The ZC LLRs L_k^{zc} in (6) and (7) are shown with dashed lines in Fig. 2(a). LLR L_1 is approximated by a single line, while LLR L_2 is approximated by a piecewise linear function with two lines, since there are two ZC points. The ZC approximations are very close to the exact LLRs (4) around the ZC points (black markers in Fig. 2(a)). In the next section, we analyze the performance of the SD FEC decoder using the three LLR calculations discussed, i.e., L_k , L_k^{awgn} , and L_k^{zc} .

Numerical Results

We use Monte Carlo simulations for the system in Fig. 1 with the simulation parameters in Tab. 1. We sweep the OMA from -10 to 5 dBm, and calculate the post-FEC BER at every point. For SD decoding, we use the Chase decoding algorithm [20] with 3 least-reliable bits and flipping up to 2 bits, resulting in 7 test patterns. The results are shown in Fig. 2(b) in three groups of curves that represent net bitrates of 400, 600, and 800 Gb/s. Note that the symbol rate, noise bandwidth, and constants p_0 and p_1 from Tab. 1 (for 400G) scale linearly with the bitrate.

Figure 2(b) shows that for all cases the BER converges to an error floor at high OMA due to the

presence of RIN in the channel. In general, the effect of RIN becomes more dominant as the data rate increases. Furthermore, there is a nonnegligible penalty caused by using the mismatched AWGN L_k^{awgn} instead of L_k as we have predicted. At high OMA, using L_k^{awgn} results in an error floor increase of up to 5 times for 400G, 3.3 times for 600G, and 3 times for 800G with respect to the exact L_k . Then, our low-complexity approximation L_k^{zc} recovers this penalty and matches the exact L_k in terms of post-FEC BER. Therefore, we conclude that there is no performance loss when using our piecewise linear approximation L_k^{zc} instead of the quadratic functions resulting from the exact L_k , as the soft input for the SD decoder.

For the system parameters, this performance recovery also translates into achieving the KP4 FEC threshold of 2.26×10^{-4} [21] with 0.1 and 0.4 dB less in terms of OMA for 400G, and 600G, respectively. For 800G, the BER of L_k^{awgn} is already in the error floor region at the KP4 threshold. This represents a case where the mismatch in LLRs can lead to a scenario in which the KP4 threshold is not met or leaves no margin of operation. However, with L_k^{zc} we operate with reasonable margin, and thus, reliable error-free communication is feasible for next-generation transceivers.

Conclusions

We studied the problem of LLR calculations for PAM-4 in next-generation RIN-dominated IM-DD systems. We showed that using mismatched AWGN LLRs results in a nonnegligible performance penalty, and proposed a low-complexity piecewise linear approximation that results in virtually no performance loss in terms of post-FEC BER. Future work includes extending the results to higher-order PAM, as well as experimental validation of the proposed technique with stronger (capacity-approaching) FEC codes.

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